

GROVE AI Open Ablation Evidence Package

Technical summary. The archived GROVE system retains a large identification advantage over the best tested single-pass baseline. The new multi-system sensitivity analysis shows that raw IoG ranking changes at strict thresholds because a Llama baseline uses broad boxes, while GROVE remains stronger under IoU, tightness, and one-to-one box+label criteria. Exact reruns requiring missing local checkpoints are labeled NOT_RUN.

Component ablation evidence

The first table reports identification and provenance. The second reports grounding behavior for the same variants. Trace-derived, proxy, evaluation-only, and NOT_RUN rows remain visibly labeled.

System Variant	Exactness Label	TP/FP/FN	Precision	Recall	Micro F1	Macro F1	95% CI	Mean Latency
Full GROVE	Exact	167/63/43	0.7261	0.7952	0.7591	0.5593	[0.7014, 0.8077]	19.937
Trace-enabled GROVE control	Exact cached control	142/206/68	0.4080	0.6762	0.5090	0.4189	[0.4481, 0.5629]	22.474
No Path 2 verification	Trace-derived	141/208/69	0.4040	0.6714	0.5045	0.4158	[0.4457, 0.5574]	
No reconciliation (union)	Trace-derived	142/209/68	0.4046	0.6762	0.5062	0.4170	[0.4457, 0.5593]	
GroundingDINO only	Trace-derived	131/172/79	0.4323	0.6238	0.5107	0.4135	[0.4524, 0.5635]	
GroundingDINO + DETR/OpenCLIP stack	Trace-derived	141/204/69	0.4087	0.6714	0.5081	0.4190	[0.4492, 0.5619]	
No fallback grounding (boxes removed)	Trace-derived	142/206/68	0.4080	0.6762	0.5090	0.4189	[0.4481, 0.5629]	
Caption-only keyword proxy	Proxy	205/1047/5	0.1637	0.9762	0.2804	0.2538	[0.2392, 0.3163]	
Image-only direct-Qwen proxy	Proxy	130/209/80	0.3835	0.6190	0.4736	0.3470	[0.4219, 0.5229]	7.132
Qwen 3.5 9B existing single-pass	Proxy	130/209/80	0.3835	0.6190	0.4736	0.3470	[0.4219, 0.5229]	7.132
No Other category (evaluation-only)	Eval-only	158/59/28	0.7281	0.8495	0.7841	0.5714	[0.7251, 0.8320]	N/A eval-only
Fallback only	NOT_RUN							
Caption-only reasoning (exact)	NOT_RUN							
Image-only modular reasoning (exact)	NOT_RUN							

System Variant	Exactness Label	TP/FP/FN	Precision	Recall	Micro F1	Macro F1	95% CI	Mean Latency
Qwen 3.5 9B matched-schema single-pass	NOT_RUN							
No compliance rules	NOT_RUN							
No Other category (prompt rerun)	NOT_RUN							

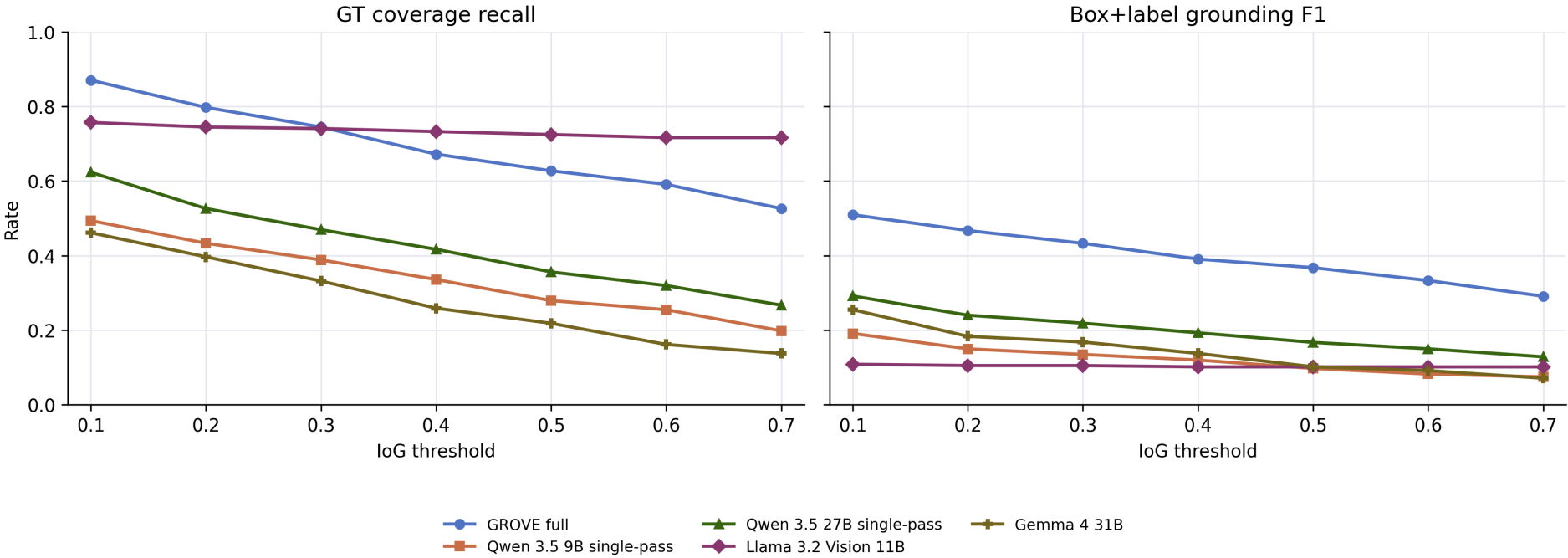
System Variant	GT Coverage	Pred Coverage	Tightness	Mean IoU	Box+Label F1	Predicted Hazards	No-box Rate
Full GROVE	0.7449	0.6218	0.7941	0.6282	0.4330	276	0.0036
Trace-enabled GROVE control	0.7530	0.4722	0.7843	0.6412	0.3273	418	0.0120
No Path 2 verification	0.7530	0.4722	0.7843	0.6412	0.3242	418	0.0120
No reconciliation (union)	0.7530	0.4723	0.7854	0.6409	0.3263	420	0.0119
GroundingDINO only	0.7328	0.4901	0.7782	0.6310	0.3389	355	0.0000
GroundingDINO + DETR/OpenCLIP stack	0.7530	0.4722	0.7843	0.6412	0.3242	413	0.0000
No fallback grounding (boxes removed)	0.7328	0.4901	0.7782	0.6310	0.3422	418	0.1507
Caption-only keyword proxy	0.0000	0.0000	0.0000	0.0000	0.0000	1252	1.0000
Image-only direct-Qwen proxy	0.3887	0.2587	0.6963	0.4725	0.1351	471	0.3928
Qwen 3.5 9B existing single-pass	0.3887	0.2587	0.6963	0.4725	0.1351	471	0.3928
No Other category (evaluation-only)	0.7568	0.6145	0.7926	0.6375	0.4463	263	0.0038
Fallback only							
Caption-only reasoning (exact)							
Image-only modular reasoning (exact)							
Qwen 3.5 9B matched-schema single-pass							
No compliance rules							
No Other category (prompt rerun)							

Grounding threshold sensitivity

Category-agnostic IoG is reported alongside IoU, IoP/tightness, and category-aware one-to-one box+label F1. The full 0--1 axis is retained.

Multi-system grounding sensitivity

203-image OSHA-aligned benchmark; full 0-1 y-axis; category-aware matches use one-to-one greedy IoG matching



IoG sensitivity

System	IoG@0.3	IoG@0.4	IoG@0.5	IoG@0.6	IoG@0.7	No-box Rate
GROVE full	0.7449	0.6721	0.6275	0.5911	0.5263	0.0036
Qwen 3.5 9B single-pass	0.3887	0.3360	0.2794	0.2551	0.1984	0.3928
Qwen 3.5 27B single-pass	0.4696	0.4170	0.3563	0.3198	0.2672	0.3220

System	IoG@0.3	IoG@0.4	IoG@0.5	IoG@0.6	IoG@0.7	No-box Rate
Llama 3.2 Vision 11B	0.7409	0.7328	0.7247	0.7166	0.7166	0.1151
Gemma 4 31B	0.3320	0.2591	0.2186	0.1619	0.1377	0.2927

IoU sensitivity

System	IoU@0.3	IoU@0.4	IoU@0.5	Mean IoU	Mean Tightness	No-box Rate
GROVE full	0.6397	0.5628	0.5101	0.7352	0.9319	0.0036
Qwen 3.5 9B single-pass	0.2955	0.1862	0.1417	0.5492	0.8140	0.3928
Qwen 3.5 27B single-pass	0.3522	0.2672	0.2267	0.5908	0.8309	0.3220
Llama 3.2 Vision 11B	0.3806	0.2915	0.1984	0.6096	0.6193	0.1151
Gemma 4 31B	0.2186	0.1741	0.1134	0.5212	0.7823	0.2927

IoP sensitivity

System	IoP@0.3	IoP@0.5	Pred Containment@0.3	Mean Tightness	No-box Rate
GROVE full	0.8219	0.7692	0.8436	0.9265	0.0036
Qwen 3.5 9B single-pass	0.5263	0.4413	0.4895	0.8585	0.3928
Qwen 3.5 27B single-pass	0.5628	0.4899	0.6484	0.8579	0.3220
Llama 3.2 Vision 11B	0.4291	0.2713	0.4056	0.6489	0.1151
Gemma 4 31B	0.4332	0.3360	0.6828	0.7926	0.2927

Box+label sensitivity

System	Box+Label F1@IoG0.3	Box+Label F1@IoG0.5	Box+Label F1@IoU0.3	Box+Label F1@IoU0.5
GROVE full	0.4330	0.3678	0.3602	0.3027
Qwen 3.5 9B single-pass	0.1351	0.0976	0.1163	0.0600
Qwen 3.5 27B single-pass	0.2189	0.1674	0.1717	0.1245
Llama 3.2 Vision 11B	0.1053	0.1018	0.0561	0.0246

System	Box+Label F1@IoG0.3	Box+Label F1@IoG0.5	Box+Label F1@IoU0.3	Box+Label F1@IoU0.5
Gemma 4 31B	0.1684	0.1020	0.1173	0.0663

Compliance, taxonomy, and latency checks

No-hazard false-positive image rate

System	status	n_no_hazard_images	fp_images	no_hazard_fp_image_rate	no_hazard_specificity	not_run_reason
Full GROVE	COMPUTED	98	13	0.1327	0.8673	
GROVE-NoComplianceRules	NOT_RUN	98				Required qwen3.5:9b model is absent from local Ollama manifests; the Ollama server is not running and internet/model downloads are disallowed.
Qwen 3.5 9B single-pass	COMPUTED	98	65	0.6633	0.3367	
Best single-pass baseline	COMPUTED	98	24	0.2449	0.7551	

No-Other evaluation

System	status	exactness	TP	FP	FN	precision	recall	micro_f1	macro_f1	f1_ci95	gt_coverage_iog03	no_box_rate	interpretation
Full GROVE (7 categories)	COMPUTED	Exact	167	63	43	0.7261	0.7952	0.7591	0.5593	[0.7014, 0.8077]	0.7449	0.0036	Primary seven-category result.
GROVE-NoOther-EvalOnly	COMPUTED	Eval-only	158	59	28	0.7281	0.8495	0.7841	0.5714	[0.7251, 0.8320]	0.7568	0.0038	Six-category evaluation only; does not model category redistribution.
GROVE-NoOther-PromptRerun	NOT_RUN	NOT_RUN											Exact prompt-level effect remains unmeasured.

Latency on cached original runs

system	status	mean_latency_sec	median_latency_sec	p90_latency_sec	p95_latency_sec	std_latency_sec	valid_timing_images	missing_timing_images
GROVE_full_paper_archived	COMPUTED	19.937	19.980	24.170	24.720	2.748	203	0
baseline_direct_gemma4_31b	COMPUTED	4.890	3.110	11.190	12.060	3.814	203	0
baseline_direct_llama32_vision_11b_apr2026	COMPUTED	8.658	7.990	11.180	12.740	1.981	203	0
baseline_direct_qwen35_27b	COMPUTED	7.010	7.820	13.240	13.450	4.645	203	0

system	status	mean_latency_sec	median_latency_sec	p90_latency_sec	p95_latency_sec	std_latency_sec	valid_timing_images	missing_timing_images
baseline_direct_qwen35_9b	COMPUTED	7.132	8.270	8.650	9.040	2.850	203	0

Latency reflects original execution environments and is not a controlled same-hardware benchmark. Trace-derived and evaluation-only variants do not receive inferred runtimes.

Claim traceability

claim	evidence	result	caveat
Full GROVE outperforms the best tested single-pass baseline on identification.	table_2_paired_tests.csv; table_1_component_ablation_completed.csv	Paired identification F1 difference +0.2047, 95% CI [0.1353, 0.2728].	This supports the tested benchmark comparison, not a universal single-pass ceiling.
Path 2 and reconciliation provide measurable but modest trace-level gains.	table_1_component_ablation_completed.csv	Matched trace control F1=0.5090; no Path 2 F1=0.5045 (delta -0.0045); no reconciliation F1=0.5062 (delta -0.0028).	These comparisons use the trace-enabled control, not archived full-system internals.
The cached grounding gain is not explained solely by fallback boxes.	table_1_component_ablation_completed.csv	Matched trace-control GT coverage=0.7530; GroundingDINO-only=0.7328; removing fallback boxes=0.7328.	A true fallback-only primary-grounder rerun is NOT_RUN because its local weights are unavailable.
Raw IoG ranking is not invariant to threshold.	grounding_sensitivity_multisystem/iog_sensitivity.csv	GROVE IoG@0.5=0.6275; Llama IoG@0.5=0.7247.	High IoG can result from broad predicted boxes.
GROVE is stronger under stricter localization and box+label criteria.	iou_sensitivity.csv; iop_sensitivity.csv; box_label_sensitivity.csv	GROVE IoU@0.5=0.5101 and box+label F1@IoU0.5=0.3027.	Strict IoU may penalize semantically valid relational boxes.
Exact causal claims for fallback-only, caption-only, image-only, compliance rules, and matched schema remain unresolved.	table_1_component_ablation_completed.csv; completion_report.md	Required exact reruns are explicitly marked NOT_RUN.	Proxy rows are retained only as labeled alternatives and are not treated as exact evidence.

Methods and limitations

All computed evidence uses the same 203-image universe, local COCO labels, canonical taxonomy, and deterministic seed. IoG, IoU, and IoP use their standard area definitions. Box+label metrics require category equality and one-to-one greedy matching by descending overlap.

Two categories have sparse support, the Other class is broad, and exact inference ablations were blocked by missing local model checkpoints. No numbers were imputed for blocked variants. The results support a benchmark-specific modularity claim but not a universal ceiling for single-pass VLMs.